Ctrl+ArcZ

Screened before it is signed. Recallable until it is claimed. Repeatable without handing over your wallet.

USDC payments on Arc with the three things a plain transfer does not have: a firewall that refuses a bad recipient before anything is signed, a lock the sender can undo until the recipient proves the money was meant for them, and a bounded spend box that lets a merchant or an agent charge you again without ever touching your wallet.

ARC TESTNET · LIVE | CHAIN ID 5042002

`0x8dAb7148cdc31DAcad6d7e12161AA3DEDb572Dca`

ANDROID **Google Play** | WEB **ctrlarcz.xyz** | SOURCE **Farukest/Ctrl-ArcZ** | PACKAGE **@ctrl-arcz/sdk** | IOS **Coming soon**

THE PROBLEM

A plain transfer is final, blind and one-shot

Final

Signed is settled. There is no undo on a plain transfer, for the right recipient or the wrong one.

Blind

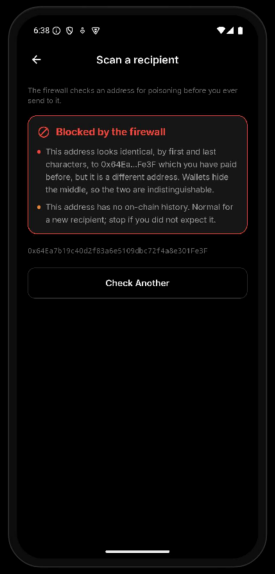
Nobody reads the middle of an address. Poisoning lives here: the attacker grinds a lookalike of one you already paid, and the **$1 test transfer** confirms perfectly — to the wrong one too.

`0x64Ea…Fe3F`

the one you pay

`0x64Ea…Fe3F`

the one the attacker ground



Escrow alone would lock the money for the attacker. The firewall refuses before a send exists.

One-shot

A payment that must repeat rests on an unlimited approve or a shared key — the merchant, or the agent, is handed more than the payment.

THE SOLUTION

Four layers, one contract, no custody

Layer 1

Firewall

A graded verdict before anything is signed. **sendProtected** runs the scan itself, so installing the SDK is what makes a send protected.

Layer 2

Protected transfer

USDC sits in the contract until the recipient proves a code. The sender can cancel; expiry refunds automatically.

Layer 3

Clean history

Zero-value bait and unknown tokens drop out of the list, and every settled claim feeds back into Layer 1.

Layer 4

Payer shield

The paying side is shielded too: a spend box owned by a fresh **stealth address**, a co-signer on every draw, and an agent key bounded by the contract.

Not another wallet. One deployment, many tenants: an integrator registers a config and gets its own recall window, claim mode and minimum worth protecting.

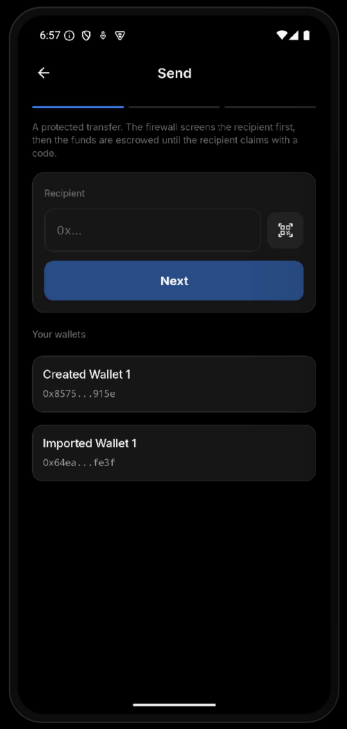
LAYER 1 IN THE FLOW

The send, split into decisions

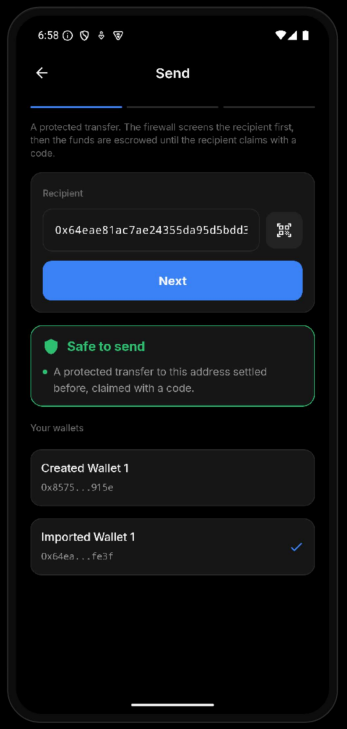
01 Step one is only the recipient. The firewall screens it **before the amount exists**.

02 This one reads **safe** because a transfer here already settled with a code, Layer 3 feeding Layer 1.

> **Safe to send**
> A protected transfer to this address settled before, claimed with a code.



Recipient first, nothing else.



A reasoned verdict, not a green tick.

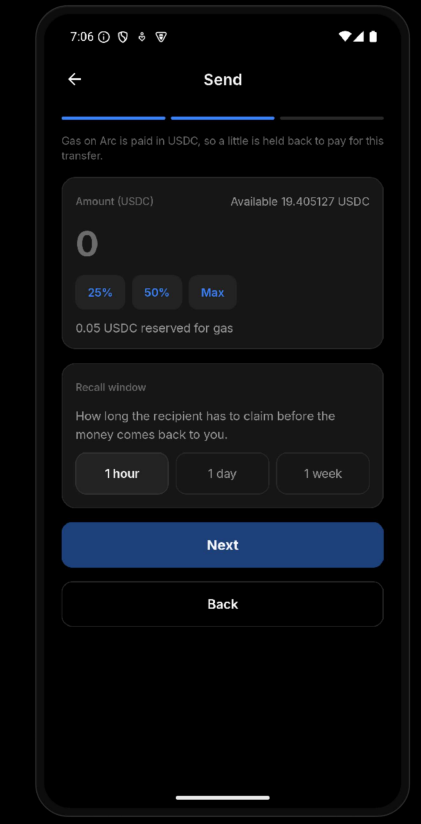
LAYER 2 IN THE FLOW

Amount and gas on Arc

On Arc, gas is paid in USDC, so the amount step has to be honest about it.

Reserved for gas	0.05 USDC
Recall window	1 hour · 1 day · 1 week
If unclaimed	returns to the sender

Confirm restates the recipient, the verdict and the amount while nothing has moved yet.



Max is not the whole balance, and the app says why.

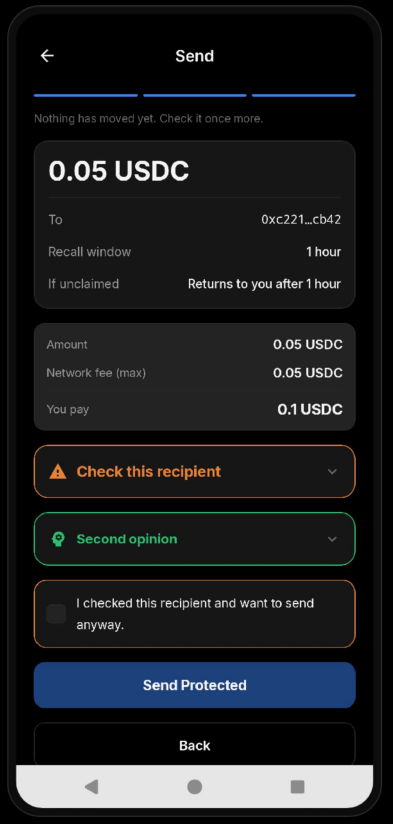
CONFIRM

Every screen that spends says what it will cost

On Arc, gas is USDC too. An amount without its fee is a lie.

01 Confirm ends in three lines — **Amount**, **Network fee (max)**, and under the rule the one number that is true: **You pay**.

02 The same block closes a send, a private payment and a bridge; a subscription states its own — total, end date, balance — before it starts charging.



Amount, fee, then the number that is true.

GRADED VERDICTS

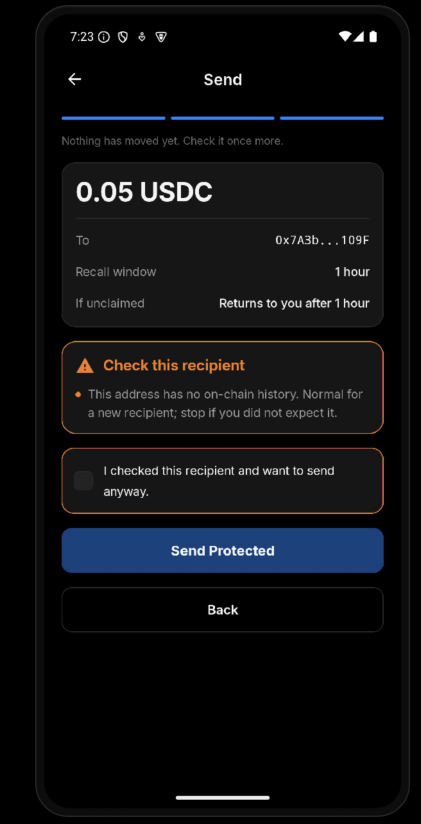
The warning gate

01 An address with no on-chain history gets a **warning**, not a block.

02 Send Protected stays disabled until the warning is accepted by hand. It cannot be clicked through out of habit.

03 A positive signal never overrides a block, and if the data source is down the firewall **fails closed**.

> **Check this recipient**
> This address has no on-chain history. Normal for a new recipient; stop if you did not expect it.



The action is visible and locked until accepted.

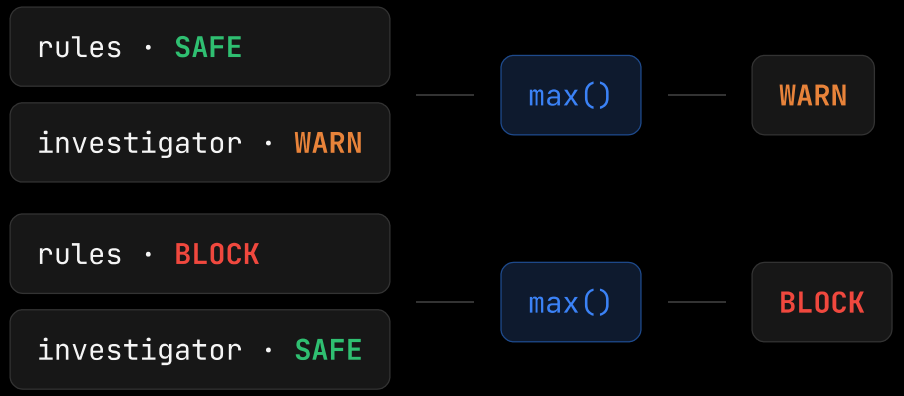
LAYER 1, EXTENDED

An investigator that can only tighten

`POST /api/investigate` gathers the signals the rule engine cannot combine. Its answer moves a verdict in one direction only.

01 Every reply is clamped to the rule engine's decision with **max**: a wrong or prompt-injected answer can refuse a good payment — it cannot approve a bad one, and it cannot clear a lookalike.

02 No key, a timeout, a malformed reply: the app returns to its **rule-only self**. The feature disappears; nothing else degrades.



A SAFE answer never lifts anything.

THE SECOND FACTOR

One code, handed over once

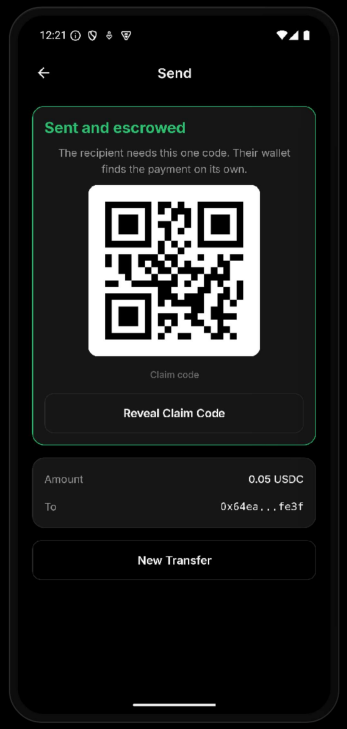
Q6MX 9CD8 PZ40 6DG2

01 Sixteen characters, **80 bits**, Crockford base32. It has to survive an offline brute force, because the recorded recipient may be the attacker.

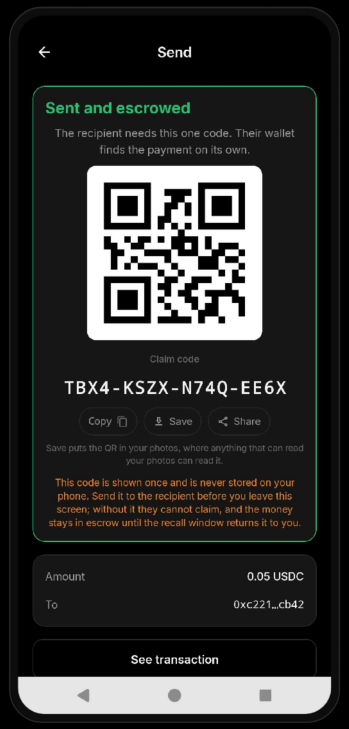
02 No link, no transfer id, no salt travels with it. The chain only ever sees the hash.

03 The reveal screen is **FLAG_SECURE**: no screenshot, no recording.

04 Under the code: **Copy, Save, Share** — and Save says it plainly: “Save puts the QR in your photos, where anything that can read your photos can read it.”



Money is in the contract; the code stays masked.



Revealed deliberately, once, to a person.

THE OTHER SIDE

Notifications without a server

No watcher service. The app reads the contract's events straight from the chain: every 15 seconds while it is open, and on a WorkManager job when it is not. Notifications are raised on the device.

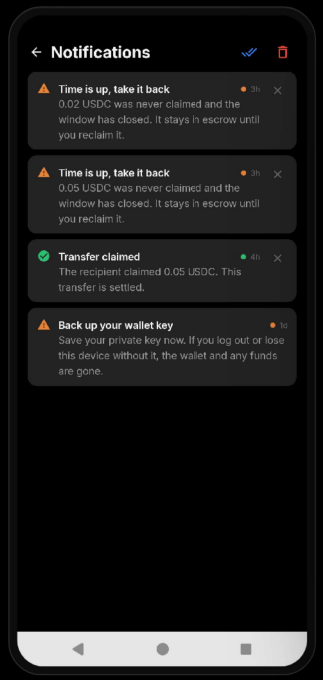
01 Filter is **TransferCreated** with the third indexed topic `to == you`. No push service, no Expo, nothing to register with.

02 Two tempos: a **15 second loop** while the app is open, and a background job when it is not — one shot on wallet connect, then every 15 minutes. The platform will not schedule that job more often, which is why the foreground loop exists.

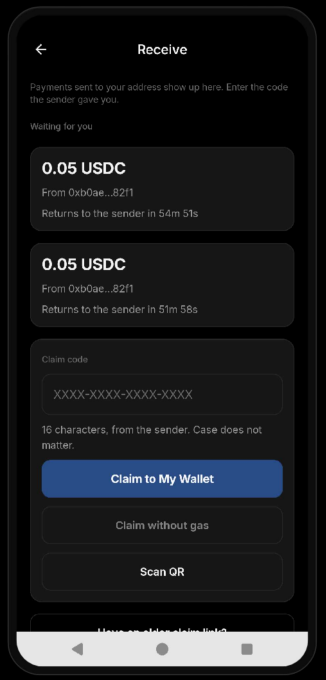
03 Delivery is **idempotent**. The cursor lives on the device and does not advance for an event it could not deliver, so the event is caught on the next pass.

04 Written twice: a persistent in-app record, plus a system banner.

05 Scope, stated plainly: **protected transfers only**. Incoming Sends, and claim, expire or recall on the ones you sent. Private Pay and Gateway arrivals are ordinary ERC-20 movements and raise nothing.



Raised on the device, from the event.



Straight into Receive.

DISCOVERY

Discovery is served, not delivered

`GET /api/announcements` — `same bytes, every caller` — `viewing key · in your browser`

01 The endpoint takes **no address**. Which announcement is yours is decided on your device, with the viewing key — it is never asked, so it is never known.

02 The notification is a convenience, not the mechanism: Receive **reads the chain itself**, so a pending transfer shows up even with notifications off.

03 **One request** where there used to be a 219-part one — and neither the server nor an explorer learns who asked.

SETTLEMENT

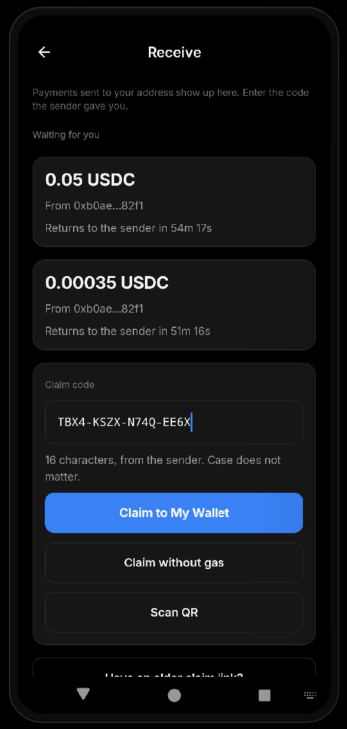
Claim without gas

01 No link is sent. The recipient's wallet finds the pending payment **from its own address**.

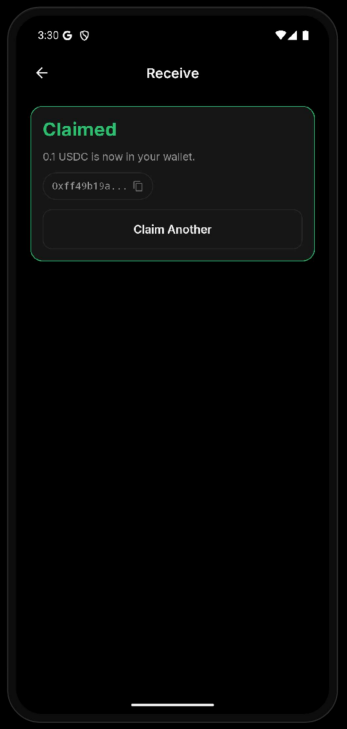
02 A wallet holding **no USDC at all** claims anyway: the claim rides an **EntryPoint v0.7** user operation and Circle Gas Station's paymaster pays the gas. The relayer's balance does not move.

03 Measured: a Circle Smart Account holding nothing claimed; the paymaster paid **0.0062 USDC**, the relayer's delta was **0**, and the recipient's nonce stayed 0.

04 Cancel is the sender's at any time; after the window, **reclaimExpired** refunds, only ever to the sender.



Found, not linked to.



Claimed gaslessly, with the hash shown.

CONTRACT BEHAVIOUR

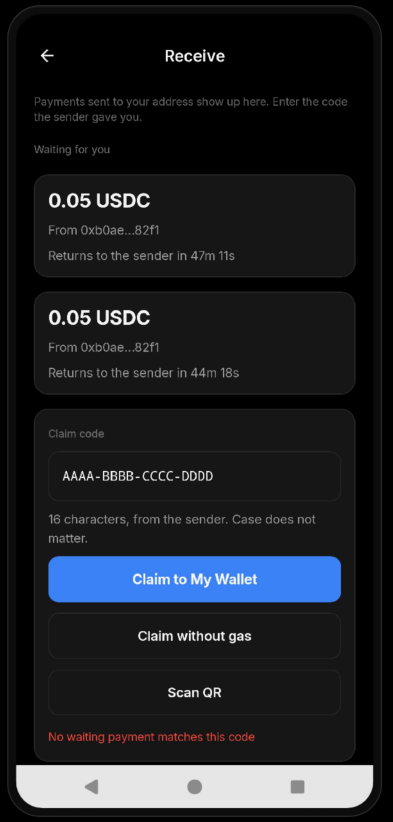
Wrong code, no cost

A wrong code returns false. It does not revert.

01 An attempt limiter cannot be built on a reverting call, because the revert would roll back the counter recording the guess.

02 Five wrong guesses freeze the transfer; the sender can still cancel and re-send.

03 A mistyped code in the app **burns no on-chain attempt**. The five stay for a real attacker.



Rejected in the app, attempts untouched.

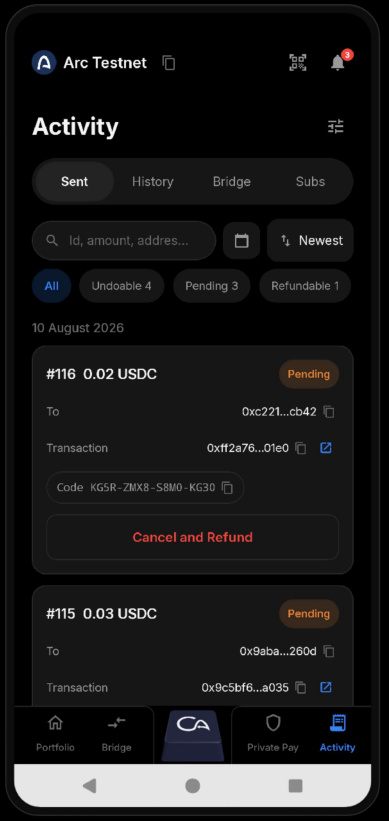
LAYER 3

History that stays readable

01 Zero-value bait transfers and unknown tokens are **filtered out of the list**, so the poisoned address never enters your history to be copied later.

02 Activity keeps the claim code with the transfer it belongs to, so a sender can hand it over again, and searches by **id, amount, address start or end, or code**.

03 Every settled claim becomes a positive signal for the firewall on the next send.



Pending, refunded, claimed, each with its code and hash.

FOR INTEGRATORS

A few calls, and the firewall runs inside them

You cannot forget to call the firewall. **sendProtected** runs it before it will sign anything.

01 A blocked recipient throws **RiskBlockedError** with the report and its reasons attached, so the integrator renders the verdict instead of inventing one.

02 **219 exported names** in `@ctrl-arcz/sdk` — 171 of them live at runtime, 112 functions and 59 values: send, claim, cancel, reclaim, scan, subscriptions, bridge.

03 **346 unit tests** on the SDK alone; **528** across the repo — 99 Foundry, 346 SDK, 50 demo-kit, 33 keeper.

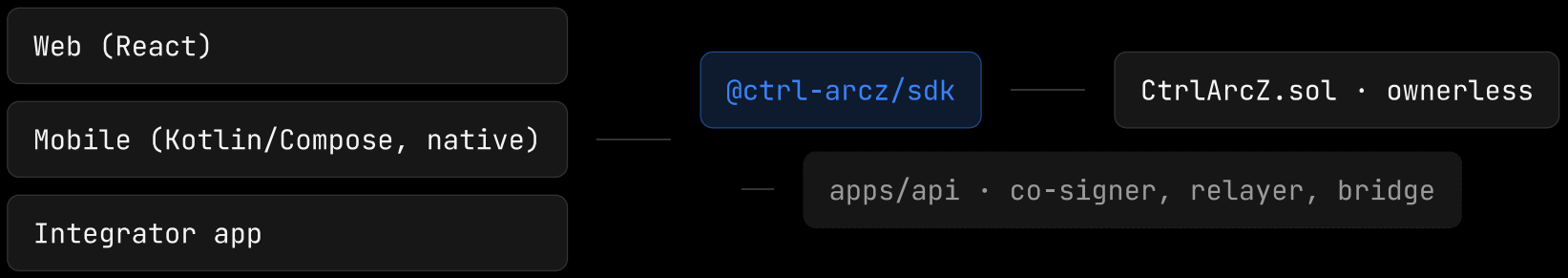
```
import { defineConfig, registerConfig,
  generateClaimCode, approveUsdc,
  sendProtected, RiskBlockedError }
  from '@ctrl-arcz/sdk'

const { configId } = await registerConfig(
  clients, defineConfig({ recallWindow: 3600 }))
const secret = generateClaimCode()
await approveUsdc(clients, amount)

try {
  const { transferId } = await sendProtected(
    clients,
    { configId, to, amount, claimHash: secret.claimHash },
    { config },
  )
} catch (err) {
  if (err instanceof RiskBlockedError)
    showVerdict(err.report)
}
```

ARCHITECTURE

Three clients, one SDK, one contract

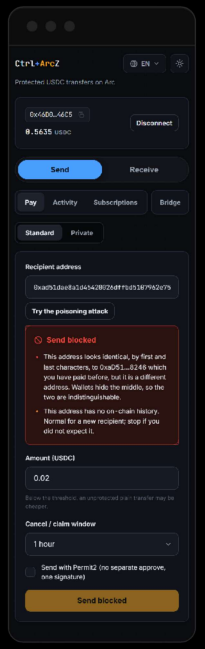


Android is native Kotlin/Compose against the same `apps/api` and the same contract. What keeps the two apps in line is `packages/sdk/parity-vectors.json`: both test suites run against it, so a port that drifts fails a test, not a payment.

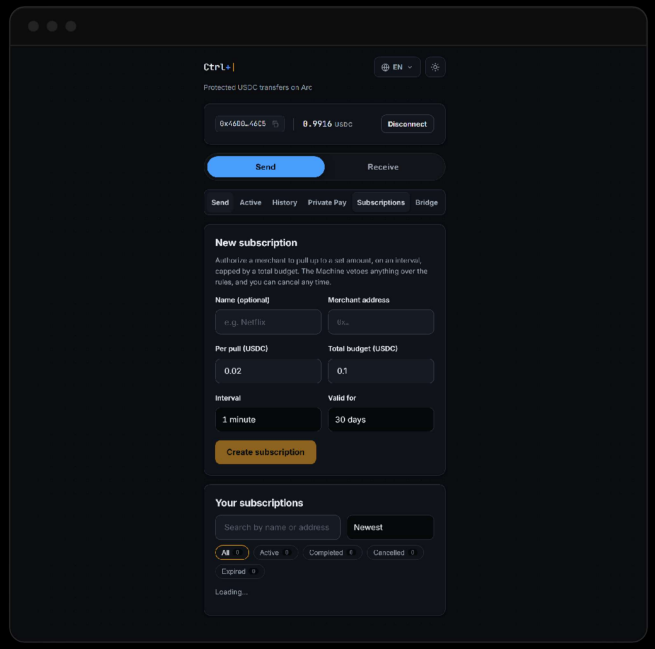
The contract holds the rules and the money. The API only pays gas, co-signs and relays; it cannot move a locked transfer, and neither can we.

THE WEB CLIENT

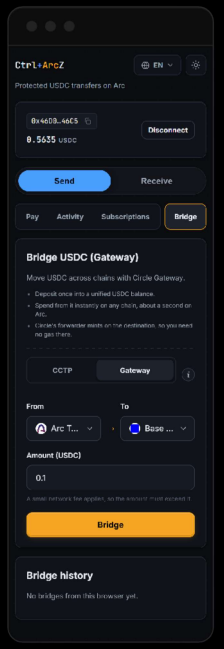
The same rules, rendered in a browser



Send blocked, two reasons — the override is a sentence you must mean. You pay 0.1.



New subscription: Netflix picked with its logo, the address label amber, Looks safe — You pay 2.434616.



Gateway, Arc Testnet to Base Sepolia: Circle fee 0.110896, You pay 1.110896.

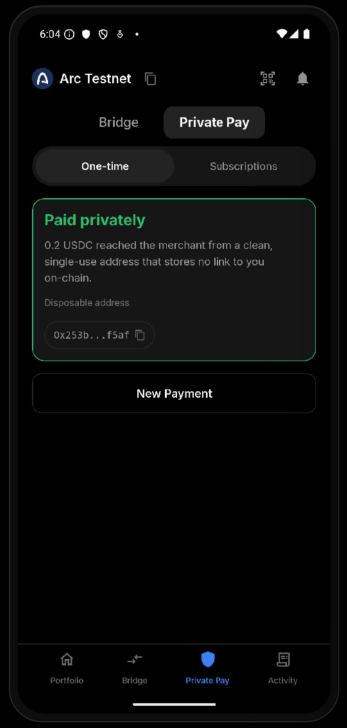
BEYOND ONE PAYMENT

Subscriptions and spend boxes

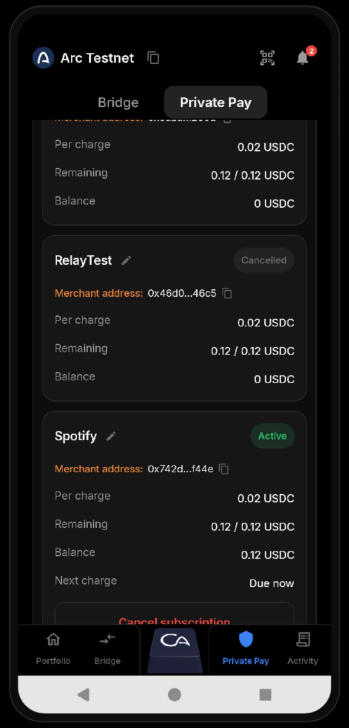
01 A one-time box pays a merchant without your address reaching them. The box is owned by a fresh **ERC-5564** stealth address.

02 The same box in **PULL mode** is a subscription: cap per pull, interval, total budget and remaining, all read from chain.

03 An **agent wallet** is the same box with a machine holding the key: it may draw up to the cap, on the interval, until the budget runs out, and the owner cancels in one transaction. The limit is enforced by the contract, not by the agent's good behaviour.



Paid from a box, not from you.



Merchant by name; per charge, remaining and balance from chain.

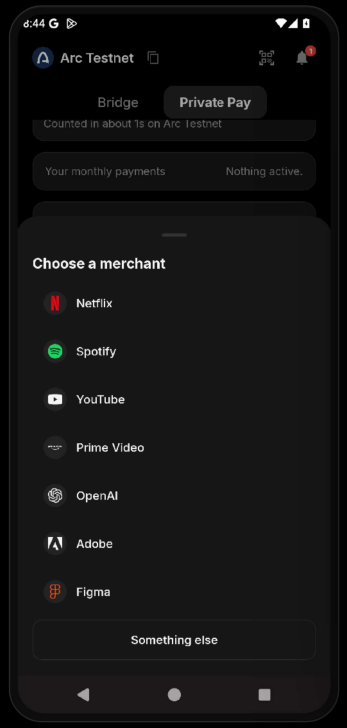
SUBSCRIPTIONS

The merchant is picked, not typed

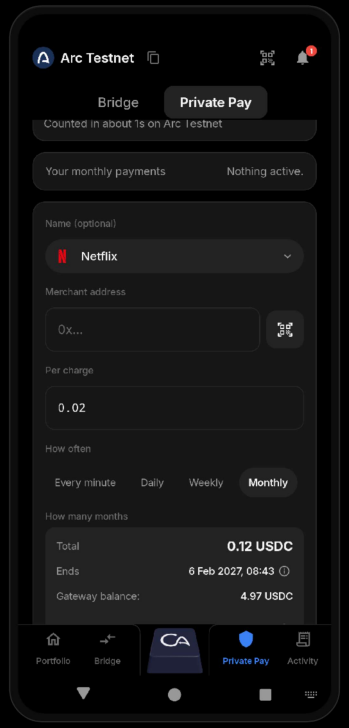
01 Not cosmetic: the name is **packed into the box's stealth announcement**, so a typo would follow the subscription onto every device that reads it.

02 Netflix, Spotify, OpenAI and the rest sit in the picker; **Something else** remains for the merchant we did not list.

03 The address field's label turned **amber**: it is the one field the firewall judges.



Picked from a list; “Something else” for the rest.



Logo and name in the pill, the summary underneath.

LAYER 4, PROVEN

An agent bounded by the chain

The keeper runs unattended, holding its own real key. What bounds it is the contract, not its judgement.

01 It reclaimed **#50** for real. The event names the keeper as `caller` and the original sender as `sender`; the recorded recipient received nothing.

02 Then it attacked its own budget, four ways. The contract refused all four.

```
TransferReclaimed(#50)
caller  → the keeper's own key
sender  → the original sender, refunded
```

Draw ten times the cap
Refused

Forge the co-signer's signature
Refused

Sweep the box to itself
Refused

Sweep to the correct vault
Refused

PRIVACY TODAY

Five steps on chain, and none of them names the payer

STEP	WHO SENDS IT ON CHAIN	LINKS TO THE PAYER
Deploy the box	Relayer	**No**
Announce the stealth address	Relayer	**No**
Fund the box	Circle's minter · Gateway mint	**No**
Gas for the stealth address	Relayer	**No**
Sweep the remainder	Stealth address	**No**

01 **StealthAnnouncer indexes msg.sender**, so announcing from your own wallet would make the stealth address pointless. The relayer announces instead.

02 Funding arrives as a **Circle Gateway mint**: what Arc records is Circle's minter paying the box. In the old design, eight boxes out of eight in one real wallet could be tied back without a viewing key; re-measured now, **zero**. APS stays in the design — no longer as a patch for a leak.

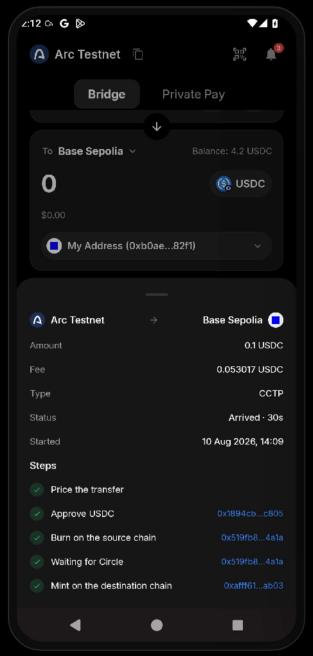
GETTING FUNDED

Bridging USDC in

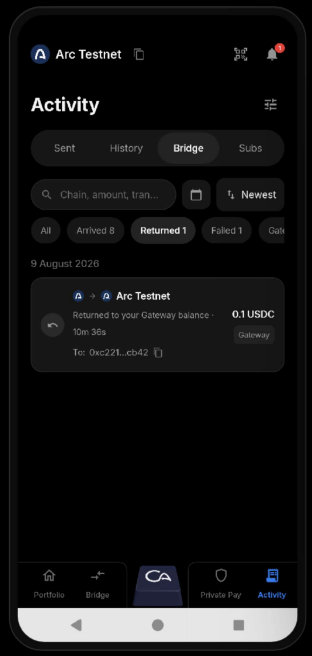
01 CCTP into Arc, five steps (price, approve, burn, attestation, mint), each one **linked to the explorer**.

02 **CCTP reaches twenty** testnet chains, **Gateway eleven**; the picker narrows to what the chosen engine really supports rather than offering a route that cannot run.

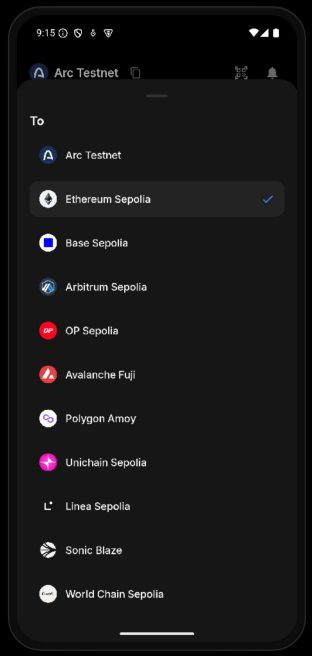
03 A failed mint **returns the money**: the row says returning, then returned. Measured twice, under ten minutes end to end, fee included.



Five steps, each one verifiable.



Returned is not failed: back in the Gateway balance in 10m 36s.



Eleven chains, real logos.

PRODUCT PHILOSOPHY

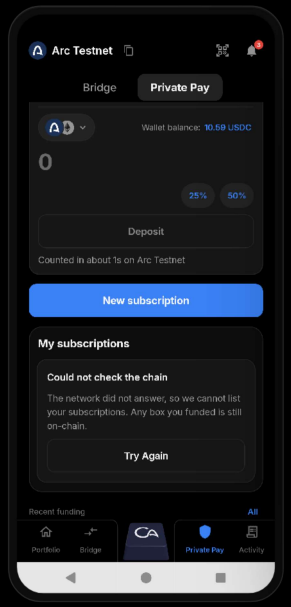
Failure, stated plainly

A payments app that rounds an unknown down to zero has taught the user to distrust it. Every unknown here is named.

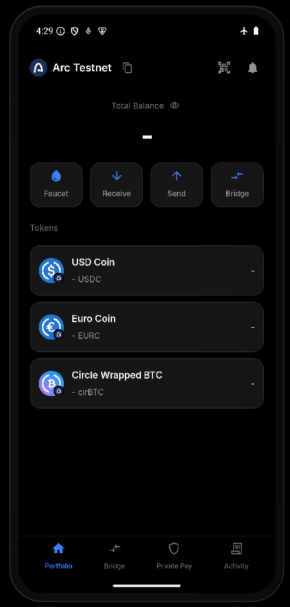
01 When the chain cannot be read it says **the network did not answer**, not "you have no subscriptions".

02 With no network the balance is **a dash**, never a fake 0.

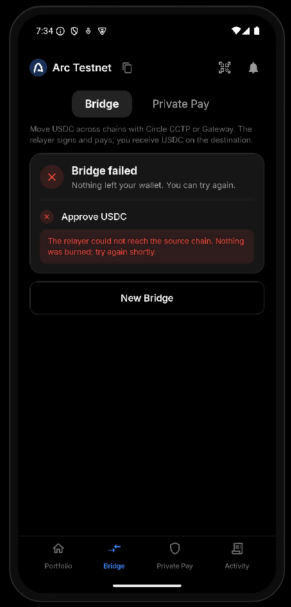
03 A failed bridge names the step, the reason, and that **nothing left your wallet**.



Unreadable, said out loud.



Offline shows a dash.



Which step, and why.

WHY THIS CHAIN

Lock-then-claim needs two transactions, and Arc makes that economical

Gas

Gas is USDC

The second transaction is cheap and predictable, and there is no separate gas token to acquire before you can move your money.

Finality

Sub-second, deterministic

The transfer settles the moment the code is entered. The recipient never watches a spinner.

Approvals

Permit2, one signature

No separate approve transaction per send.

Liquidity

CCTP and Gateway wired

Both of Circle's cross-chain routes are a single tab in the app.

LANDSCAPE

Stopping the send, without an arbiter or custody

APPROACH	STOPS THE SEND	RECOVERABLE AFTER	NEEDS AN ARBITER	TAKES CUSTODY	PLAIN P2P
Wallet address-book warning	No	No	No	No	Yes
Poisoning detection service	Warns only	No	No	No	Yes
Commerce escrow	No	By dispute	Yes	Yes	No
Circle Refund Protocol	No	By mediator	Yes	Yes	No
Ctrl+ArcZ	**Yes**	**By the sender**	**No**	**No**	**Yes**

Circle's Refund Protocol solves a different problem on purpose: it is a commerce escrow built around an arbiter. Adding an arbiter here would break the one property that makes a protected-transfer contract worth trusting.

TRUST

What is proven

The contract is ownerless: no owner, no pause, no upgrade path.

99

Foundry tests on the contract

346

SDK unit tests — 528 in total with demo-kit and keeper suites

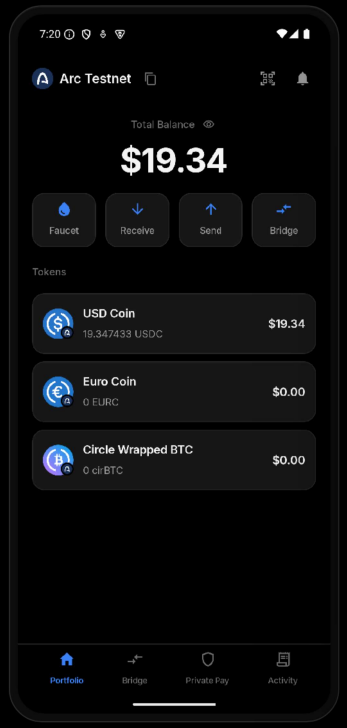
0

admin functions that can touch a locked transfer

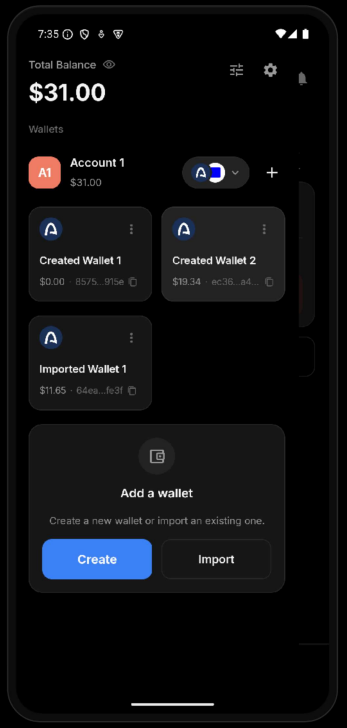
live

on Arc Testnet, testnet only, unaudited

CTRLARCZ `0x8dAb7148cdc31DAcad6d7e12161AA3DEDb572Dca`



Every number on these screens was read from Arc Testnet.



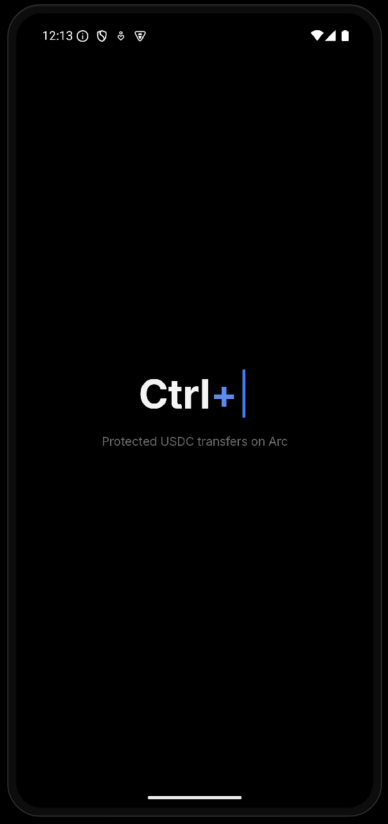
Several wallets, created and imported.

Ctrl+ArcZ

A wrong send should be refusable, and a right one should be reversible until it is claimed.

The undo shortcut, typed, erased and retyped as the product name.

ARC TESTNET · LIVE `0x8dAb…2Dca`



Ctrl+Z becoming Ctrl+ArcZ.